Lecture 1: Inverse Functions and Their Derivatives

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Determine if an inverse function exists
- ☐ Find the inverse of a function.
- ☐ Find the derivative of the inverse function.

Preliminary Examples

Example 1: Let $f(x) = 2x + 1$ with the domain $(-\infty, \infty)$, we wish to find a function that "undoes" f. Or in another words, we set $y = f(x) = 2x + 1$, we want to solve for x in terms of y.

Preliminary Examples

Example 2: Let $f(x) = x^2$ with the domain $(-\infty, \infty)$, and we wish to find a function that "undoes" f. Similarly, we set $y = f(x) = x^2$ and we try to solve for x in terms of y.

One-to-one Functions

A function is said to be one-to-one (injective) if $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ (or in other words, $f(x_1) = f(x_2)$ implies $x_1 = x_2$) for all x_1, x_2 in the domain of f.

Example: $f(x) = x + 1$ is one-to-one, but $g(x) = x^2 + 1$ is not one-to-one.

One-to-one Functions

Horizontal Line Test

A function is said to be one-to-one if and only if its graph intersect each horizontal line at most once.

Example: Consider the graphs $f(x) = x^3$, $g(x) = x^2$, $h(x) = \sqrt{x}$ and $k(x) = \sin x$. Which ones pass the horizontal line test?

One-to-one Functions

If a function f is either strictly increasing or strictly decreasing on its domain, then f is one-to-one.

Be aware, the converse is not true, there is one-to-one function that is neither strictly increasing nor strictly decreasing.

One-to-one Functions

Not all functions are one-to-one on their entire natural domain. However, by restricting the function to a smaller domain, we can create a function that is one-to-one.

Example:

$f(x) = x^2$ $\qquad\qquad$ $g(x) = \sin x$

Inverse Functions

Suppose that f is a one-to-one function on a domain D with range R. Then there exists a unique function g satisfying

$$g(f(x)) = x \text{ for all } x \text{ in the domain of } f, \text{ and}$$
$$f(g(x)) = x \text{ for all } x \text{ in the range of } f$$

Furthermore, $\text{Domain}(g) = \text{Range}(f)$ and $\text{Range}(g) = \text{Domain}(f)$ and g is also one-to-one.

This unique function g is called the inverse function of f and is denoted by f^{-1}.

Only one-to-one function can have an inverse.

If f is strictly increasing or strictly decreasing, then f is one-to-one and has an inverse.

Finding Inverses

$$f^{-1}(x) = y \text{ if and only if } y = f(x)$$

In another words,

$$(a, b) \in f \text{ if and only if } (b, a) \in f^{-1}$$

Finding Inverses

Example: Find the inverse of $y = \frac{1}{2}x + 1$, expressed as a function of x.

Finding Inverses

Example: Find the inverse of $y = x^2$, $x \geq 0$, expressed as a function of x.

Derivatives of Inverses of Differentiable Functions

Suppose that the derivative of an invertible function exists, under which circumstances is the inverse function also differentiable?

Consider the function $f(x) = x^3$ with the inverse $f^{-1}(x) = \sqrt[3]{x}$.

Derivatives of Inverses of Differentiable Functions

The derivative rule for inverses

Let f be a function with the domain over the interval I. Suppose that f is differentiable and $f'(x) \neq 0$ for all $x \in I$. Then f^{-1} is differentiable over its domain, $\mathrm{Range(f)}$ and the value of $(f^{-1})'$ at a point b in the domain of f^{-1} is the reciprocal of the value f' at the point $a = f^{-1}(b)$, i.e,

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\frac{df^{-1}}{dx}\Big|_{x=b} = \frac{1}{\frac{df}{dx}\Big|_{x=f^{-1}(b)}}$$

Derivatives of Inverses of Differentiable Functions

Derivatives of Inverses of Differentiable Functions

Example: $f(x) = x^2$, $x > 0$.

Derivatives of Inverses of Differentiable Functions

Example: Let $f(x) = x^3 - 2$, $x > 0$. Find the value of $\frac{df^{-1}}{dx}$ at $x = 6 = f(2)$ without finding a formula for $f^{-1}(x)$.